Reg. No. | | | | | | | | | | | [14EC63]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

ANTENNAS AND WAVE PROPAGATION

Time: 3 Hours **Maximum Marks:70**

Instructions:
Answer any FIVE full questions by choosing at least two questions from each part.

PART-A

1. a. Define the following with respect to antennas
   i. Radiation intensity
   ii. Directivity
   iii. Effective height
   iv. Beam solid angle **08**

   b. Prove that the directivity for a source with unidirectional power pattern given by $U = U_m \cos^n \theta$ can be expressed as 2(n+1). **06**

2. a. Explain the relative pattern of two isotropic sources of same amplitude and in-phase quadrature by deducing relevant expressions. **08**

   b. An antenna has a field pattern given by $E(\theta) = \cos^2 \theta \;\; for\; 0 \le \theta \le 90$. Find the half power beam width and exact beam area. **06**

3. a. Derive the expressions for field intensity, maxima direction and minima direction for linear array with n isotropic sources of equal amplitude with in-phase current. **08**

   b. Obtain the filed pattern for a linear uniform array of four isotropic point sources spaced $\lambda/2$ distance apart. The power is applied with equal amplitude and in-phase. Find HPBW and FNBW. **06**

4. a. Derive an expression for electric field components of a short dipole placed along the vertical axis. **10**

   b. If the current fed at the center of a $\lambda/2$ dipole is 2A and the radiation resistance is 74 ohms. Solve for the electric field in the direction of maximum radiation at a distance of 10 Kms from $\lambda/2$ dipole. **04**

PART-B

5. a. Derive an expression for radiation resistance of loop antenna. **08**

   b. Explain Babinet's principle with illustrations. **06**

6. a. Explain the operation of log periodic antenna with a neat diagram. **07**

   b. Explain the importance of corner reflector antenna and passive corner retroreflector antenna. **07**

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7. a. Explain different types of horn antennas. **08**

b. Write a brief note on the following types of antenna.

i. Embedded antenna

ii. Plasma antenna **06**

8. a. Define Maximum Usable Frequency. Derive an expression of f_{MUF} for curved surface of earth. **05**

b. Explain the mechanism of radio wave bending in the ionosphere. **05**

c. A VHF communication is to be established with 35W transmitter at 90M Hz. Find the distance up to which line of sight communication may be possible if the height of the transmitting and receiving antennas are 40m and 25m respectively. Also determine the field strength at the receiving end. **04**